

SOLUTION

TEAM NAME : 67

Solution: Predictive Public Health Resources Command Center.

Problem Statement: 5. AI for Better Healthcare in Rural Villages.

Team members:

- Dhawal Chaudhari :- Model (Technical Prompt Processing)
- Sashreek Gokhale :- Frontend (Styles :- ideation)
- Chandrasen Bodkhe :- Backend (Data Acquisition)

An intelligent, climate-triangulated epidemiological surveillance platform that transforms reactive rural crisis response into proactive 14-day outbreak foresight.



The Problem

The problem: Current rural healthcare relies on reactive disaster management, addressing outbreaks of malaria, dengue, and acute diarrheal diseases only after fatalities occur.

Affected Populations: Rural Indian villages facing monsoon road cutoffs and delayed medical interventions.

Current Gaps: Up to 40% of rural doctors and ANMs are absent or un-sanctioned. Furthermore, 35–50% of HMIS records are delayed or unrecorded due to paper registers and road closures.

Real-World Impact: Essential drugs, such as Artemisinin ACT and ORS-Zinc, suffer severe stockouts during peak transmission seasons.



Our Solution

The Solution: A predictive platform that anticipates village-level vulnerability, reconstructs missing health records, and prescribes optimal medical resource dispatch.

How it Works: The system ingests sparse HMIS facility reports, IMD meteorological grids, and Census infrastructure data to model realistic district topologies.

Key Innovation 1: A Sparse-Data Reconstruction Engine utilizes SoftImpute matrix completion and spatial kriging to recover missing data and provide 95% Bayesian Confidence Intervals.

Key Innovation 2: Climate-Health Spatial Triangulation uses 14-day and 28-day lagged rainfall data alongside temperature thresholds to forecast disease surges.

Key Innovation 3: A Prescriptive Counterfactual Engine solves multi-resource Knapsack/MILP optimization to efficiently allocate Mobile Medical Units (MMUs), doctors, and drug kits

Users, Beneficiaries & Stakeholders

Primary Users

Chief Medical Officers (CMOs) and District Collectors utilizing the interactive Streamlit command center.

Beneficiaries

Vulnerable rural populations protected from preventable severe complications and fatalities.

Stakeholders

District health authorities, field officers, and medical personnel responsible for deploying MMUs and emergency resources.



Impact

Impact Area	Current Situation	Expected Outcome	Validation Source
Data Visibility	35–50% unrecorded or delayed HMIS records.	33%–48% MAE improvement over baseline mean imputation.	Automated benchmark evaluations.
OUTBREAK RESPONSE	Reactive management after fatalities occur.	Proactive 14-day and 30-day ahead disease surge forecasting.	Gradient Boosted surge models.
Resource Efficiency	Supply chain breakdowns and drug stockouts.	43.5 preventable severe complications and fatalities averted in peak weeks.	Multi-resource MILP Optimizer simulation.

Prototype / Demonstration

Working Model: An interactive, browser-based Streamlit Public Health Command Center.

Interactive Map: A geospatial map featuring village markers color-coded by urgency tiers (Critical, High, Moderate, Stable) and road access indicators.

User Journey: Officials use the "What-If" Simulator with live sliders to adjust available doctors, MMUs, and drug kits

System Output: The platform generates SHAP Audit Cards to explain village

prioritization and exports an official field dispatch manifest (CSV/JSON) for immediate deployment.

Sustainability / Business Model

- **Operational Integration:** Functions as a district-level executive decision support tool for public health officials.
- **Data Autonomy:** Built entirely on open data paradigms including HMIS, NFHS-5, Mission Antyodaya, and IMD meteorological grids.
- **Actionable Translation:** Automatically synthesizes actionable administrative orders from data, lowering the barrier to entry for non-technical field workers.



Scalability & Future Potential

Geographical Expansion

Geographical Expansion:
Capable of ingesting custom user CSV or GeoJSON datasets to scale across any district topology.

Broad Deployment

Designed to support India's rural infrastructure by modeling realistic variables like tap water coverage, electricity reliability, and unpaved road cutoffs.

Algorithmic Growth

The modular Multi-Criteria Composite Resource Priority Index (RPI) can be continuously refined to include new intervention types and disease parameters

Conclusion

RuralHealth OS unites a robust FastAPI backend with an interactive GIS frontend, powered by a backend Random Forest machine learning ensemble. By forecasting 14-day disease outbreaks from climate and infrastructure data, the system transforms rural healthcare from reactive disaster management into proactive, life-saving medical resource allocation.





THANK YOU

